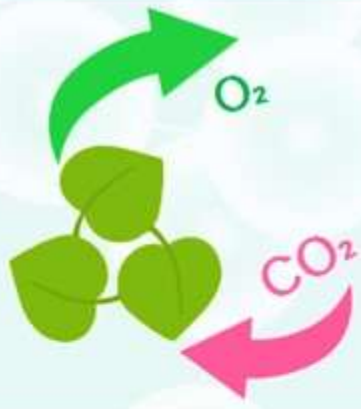


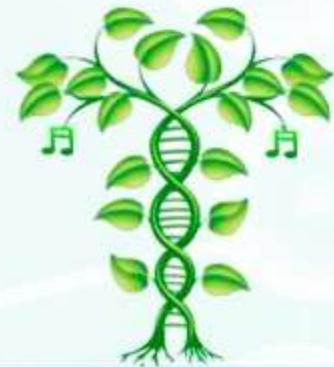


BIOLOGY



REPRODUCTION IN ANIMALS

MODULE - 12



SEXUALITY IN ANIMALS

BISEXUAL

- The organism contains both sexual organs - **male and female in one body**.
- They are also called hermaphrodite.
- Some of them don't require any partner to produce offspring.
- E.g Snails, earthworm, Echinoderms like starfish, sea cucumbers and sea urchins and some species of fish etc.



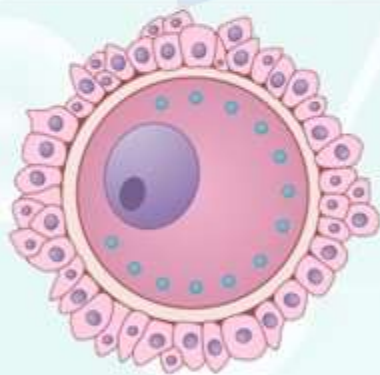
UNISEXUAL

- The organisms are divided into **two sexes** – male and female
- They have only one type of sexual organs.
- They require a partner of the opposite sex to produce offspring.
- Most vertebrate belong to this group like mammals, reptiles, amphibians, birds etc.

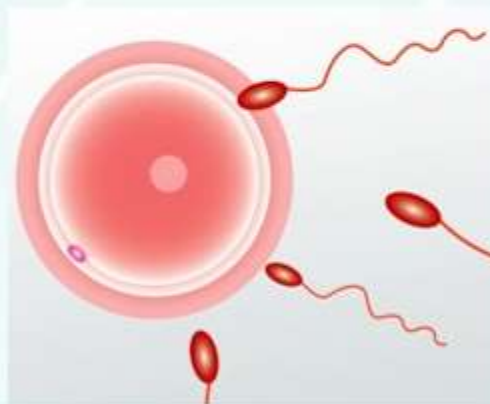


EVENTS IN SEXUAL REPRODUCTION

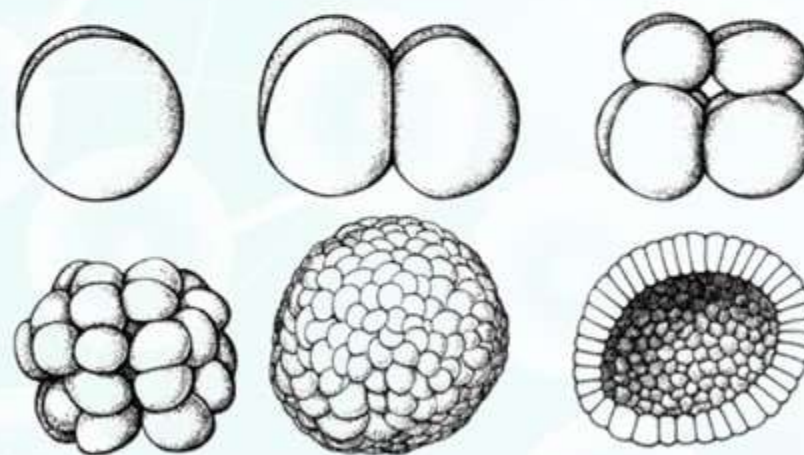
PRE-FERTILISATION



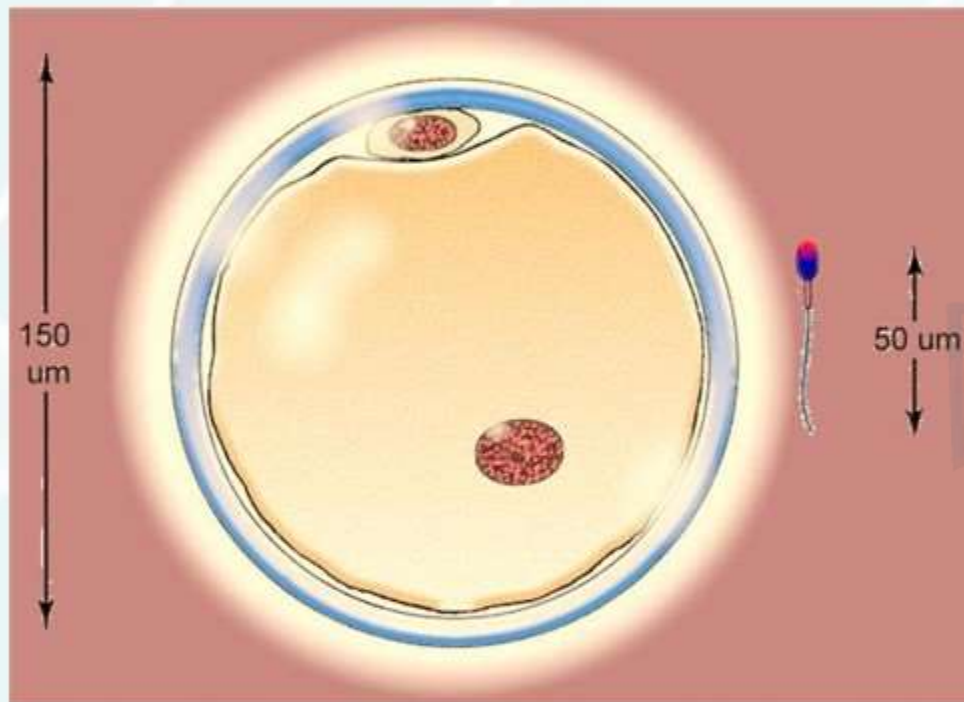
FERTILISATION



POST-FERTILISATION



PRE-FERTILISATION EVENTS



GAMETOGENESIS

The formation of gametes, haploid cells capable of carrying out sexual reproduction.

GAMETE TRANSFER

The passage of male gamete to an area where the female gamete is present in order to achieve fusion or syngamy.



TYPES OF GAMETES

HOMOGAMETE

The sex chromosomes do not differ in form, size or sex ;
Isogametes in Cladophora (alga)

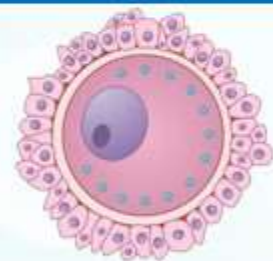


HETEROGAMETE

MALE GAMETE



FEMALE GAMETE



The male and the female gamete are different in their morphology



PRIMARY REPRODUCTIVE ORGANS IN HUMANS



MALE

- A pair of Testes
- It produces male gametes and male hormones **androgens** and **testosterone**
- Some diseases include:
 - **Testicular cancer**
 - **Bell-clapper Deformity**- testicles not attached to the scrotum
 - **Orchitis** or inflammation of the testes
 - **Cryptorchidism** or undescended testes

FEMALE



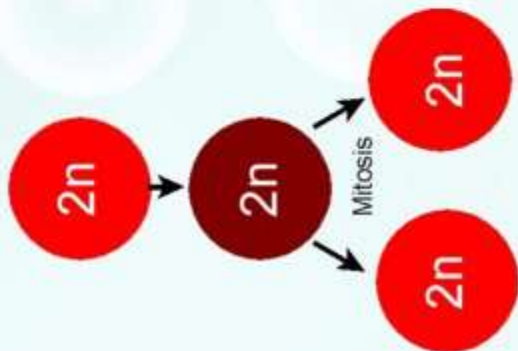
- A pair of Ovaries
- It produces egg cell and female hormones **estrogen** and **progesterone**
- Ovarian disorders include
 - **Ovarian cancer**
 - **Follicular cyst** happens when ovary does not release a mature egg
 - **Ovarian torsion** or acute lower abdominal pain
 - **Ovarian apoplexy** or rupture of the ovary
 - **Brenner Tumor**



CELL DIVISION IN GAMETOGENESIS

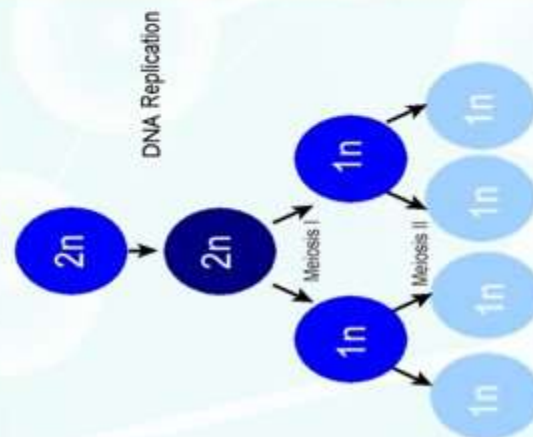
MITOSIS

- Dividing cells can be diploid or haploid.
- Occurs in both the somatic cells and the sexually reproducing cells
- The mother cells is divided into two daughter cells.
- It helps the cells to grow in number.
- It is also called equatorial division.



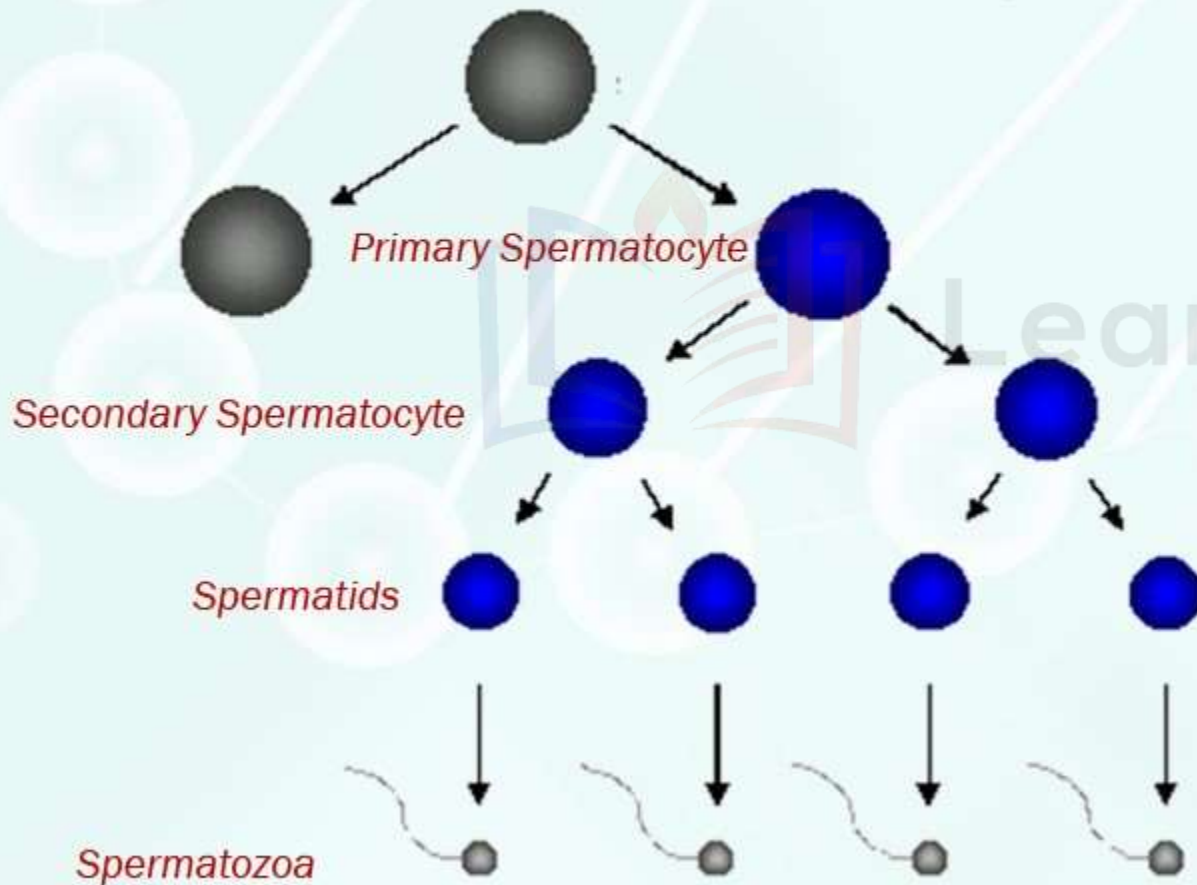
MEIOSIS

- Dividing cells are only diploid.
- Occurs only in the sexually reproductive cells or gametes.
- The mother cells is divided into 4 haploid daughter cells.
- It helps the form the haploid gametes for transmission of parental genome/DNA
- It is also called reductional division.

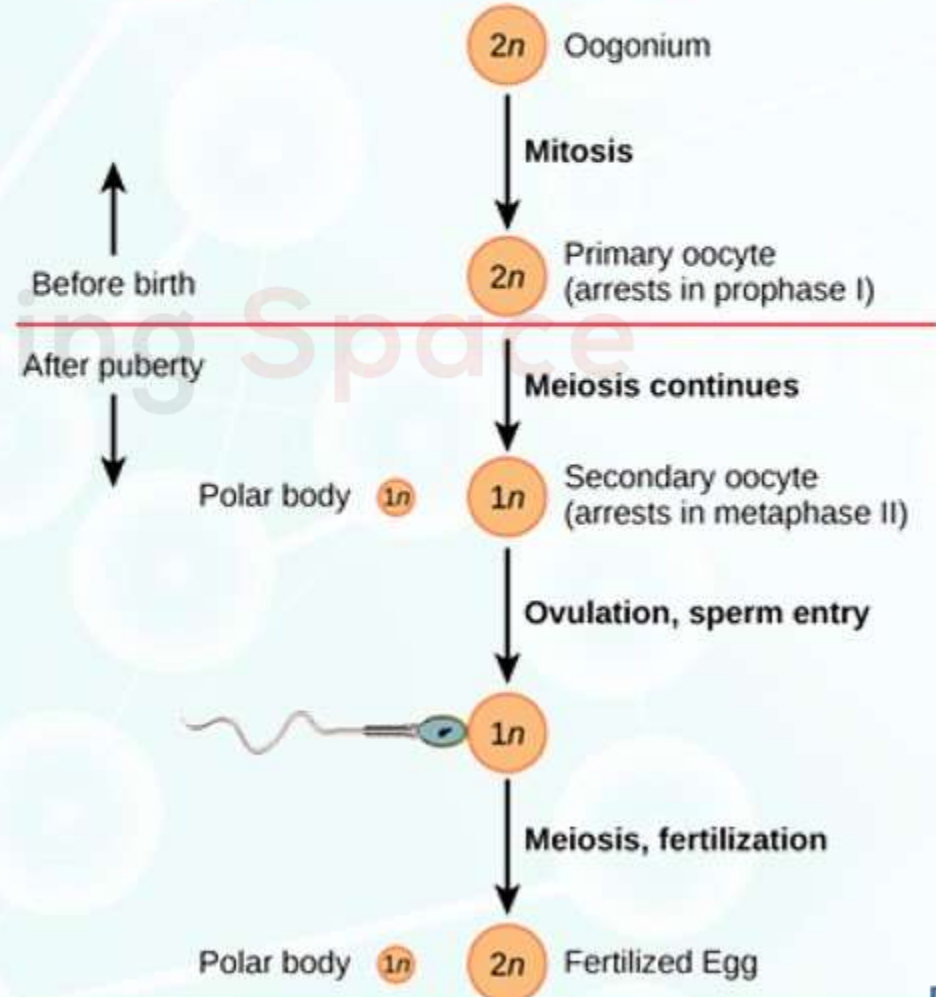


FORMATION OF GAMETES - GAMETOGENESIS

MALE

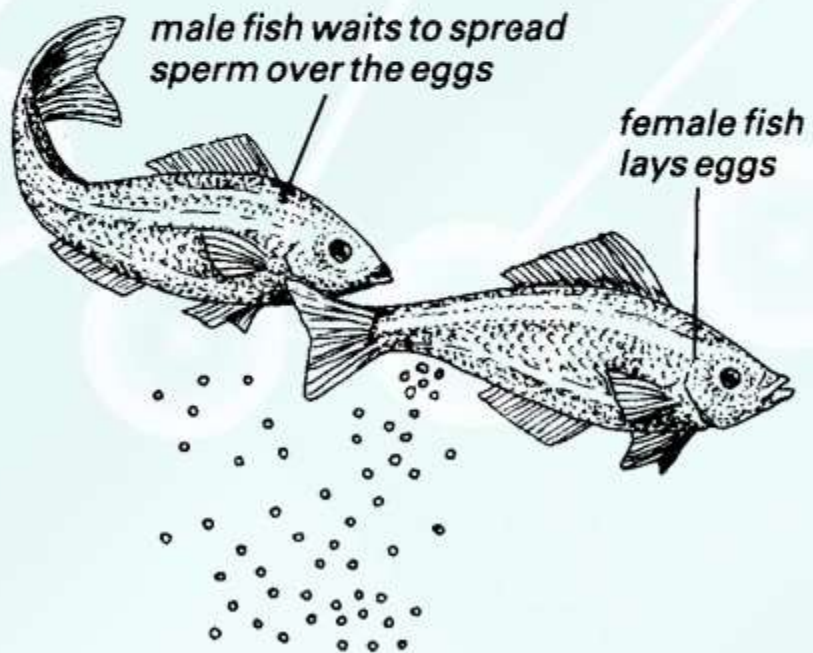


FEMALE

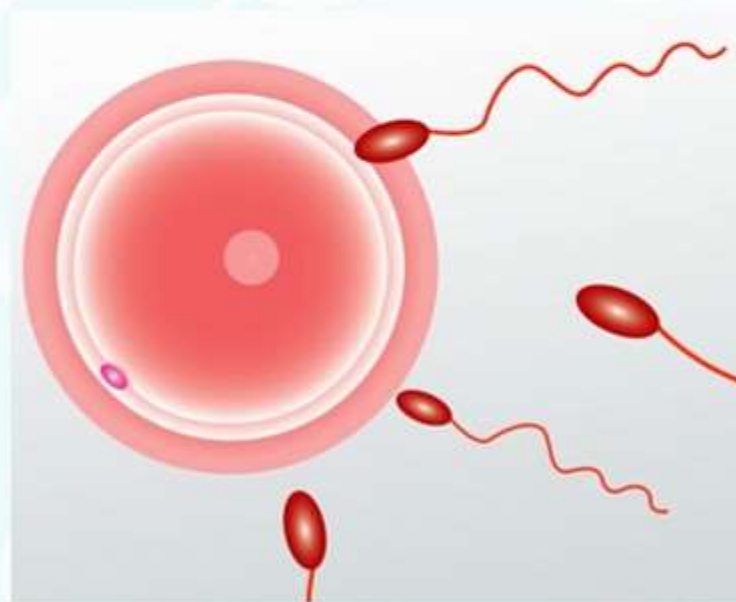


FERTILISATION

EXTERNAL FERTILISATION



INTERNAL FERTILISATION



FERTILISATION/ SYNGAMY

EXTERNAL FERTILISATION

- Syngamy occurs outside of the body of organisms.
 - Large number of both male and female gametes are released into the surrounding medium
 - zygote is formed in the external medium
- Eg. Bony fish and amphibians go through external fertilization.

INTERNAL FERTILISATION

- Syngamy occurs inside of the body of organisms.
 - Number of ova are less, but large number of male gametes are formed.
 - zygote is formed inside the body of the organism
- Eg. birds, mammals earthworm etc. go through internal fertilization.



FERTILISATION/ SYNGAMY



- Fertilization refers to the process of fusion of male and female gametes to form zygote.
- The zygote would be diploid and carry one set of DNA from each parent.
- The combination of the parental DNA is unique and no possibility of exact identical copy is possible.



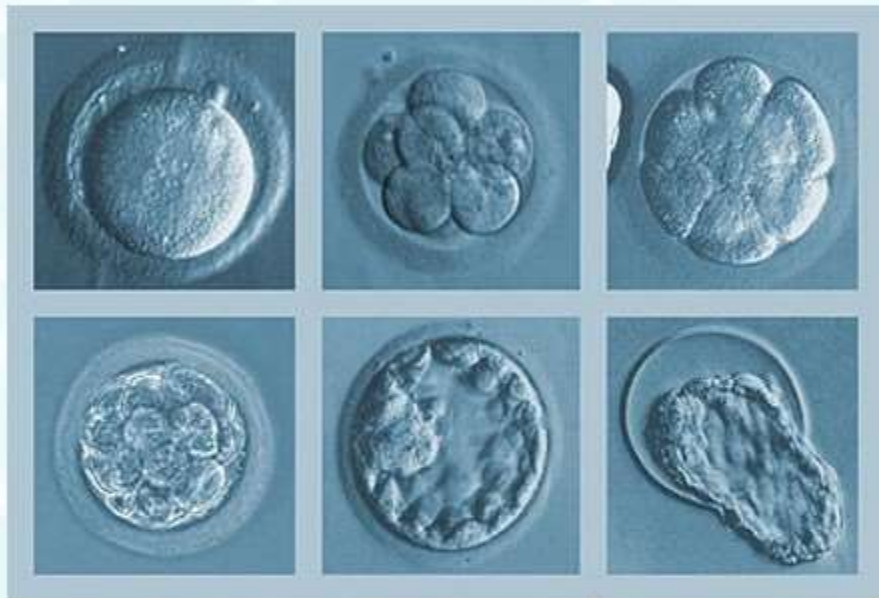
PARTHENOGENESIS



- In some species, even at the absence of fertilization, new, fully developed organisms are born from the female gamete alone.
- This is called parthenogenesis.
- Thus, parthenogenesis is the process of development of a new individual from the female gamete in the absence of fertilization.
- It is common in organisms like honey bees, some birds like turkey. It is also reported in some lizards



POST FERTILIZATION EVENTS



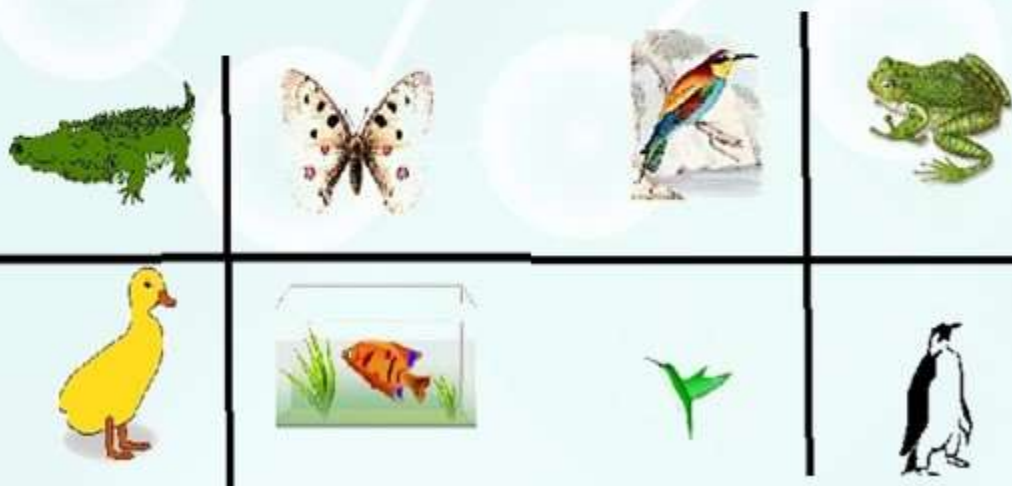
- Post-fertilization events refers to all the events of sexual reproduction taking place after fertilization, that is, after the formation of zygote.
 - The major event taking place in this stage is embryo-genesis, that is, development of the embryo.
 - As embryo-genesis continues, two processes takes place actively in the zygote.
- First one is **cell division**, that adds more cells to the zygote.
 - The second process is **cell differentiation**, that helps the new cells formed by cell division to undergo specific changes and thereby converts them into specialized tissues and organs.



POST FERTILISATION

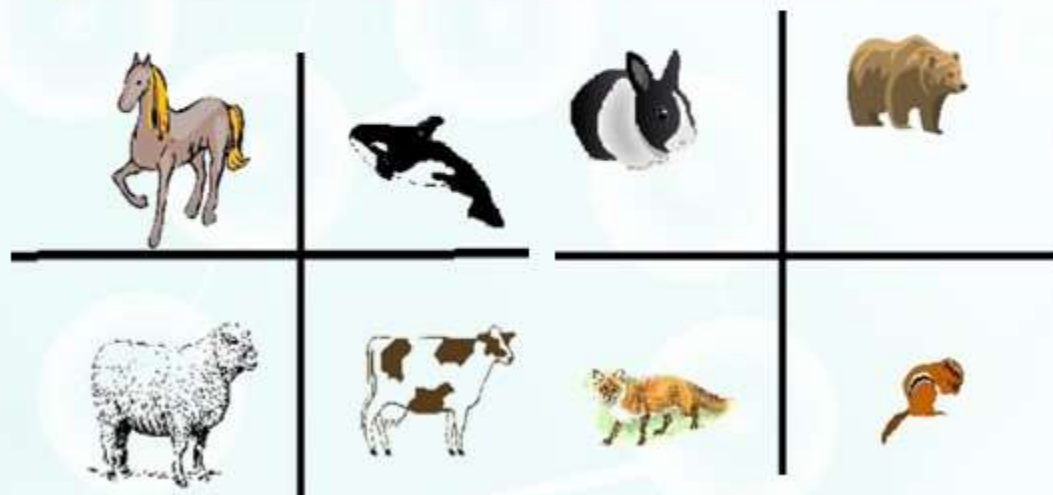
OVIPAROUS

- Animals lay fertilized or unfertilized eggs.
- Eggs have calcareous shell to protect from the harsh environment.
- Chances of survival of the offsprings are less due to threat from the predators



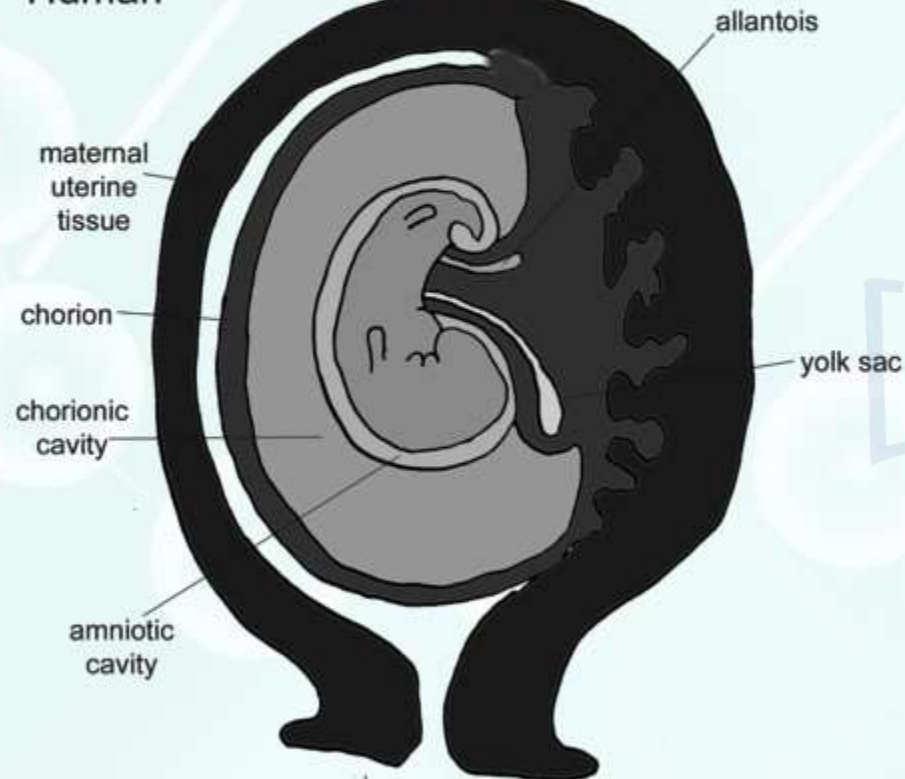
VIVIPAROUS

- Give birth to young individuals
- No shell, they are protected inside the mother's body.
- The offsprings survive more due to the parental care and protection.



PARTS OF THE EMBRYO

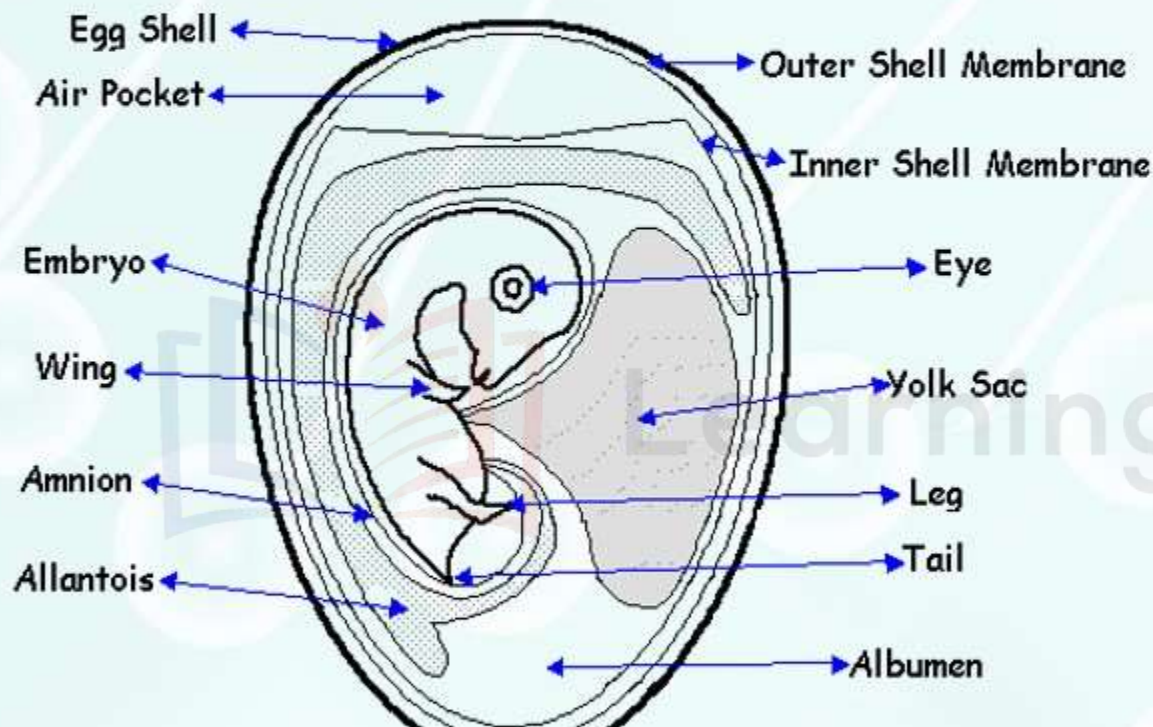
Human



- In most mammals the much dividing zygote turns into a multi celled blastula.
- It attaches itself to the uterine lining, thus stimulating the formation of a placenta,
- **Placenta serves to transfer nutrients from the mother to the growing embryo in the entire gestation.**
- By the process of gastrulation, the embryo differentiates into three types of tissue:
 - **The ectoderm**, producing the skin and nervous system;
 - **The mesoderm**, from which develop connective tissues, the circulatory system, muscles, and bones;
 - The endoderm, which forms the digestive system, lungs, and urinary system



PARTS OF THE EMBRYO



- In case of viviparous animal, nutrients pass from the placenta through the umbilical cord, and the **amnion**, a fluid-filled membrane, surrounds and protects the embryo.
- In lower animals the embryo is nourished by the **yolk**.
- The **allantois** serves as a reservoir for nitrogenous waste.
- **Albumen** or the white fluid part serves as a cushion to the growing embryo and also provides additional nutrition.



WE EXPRESS OUR SINCERE THANKS FOR VIEWING THIS VIDEO

Presented by



Learning Space

For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

OUR TEAM

G. V. Rao

Amrita Naidu

D. Sunil Kumar

L. V. Krishna

D. Chaitanya

K. Srikanth

G. Ravi Babu

D. Joji

K. Victor Babu

Visit us at:

www.learningspacedigital.com

0 8 6 6 -2 4 4 4 4 7 2

0 9 8 4 9 9 4 2 2 9 9